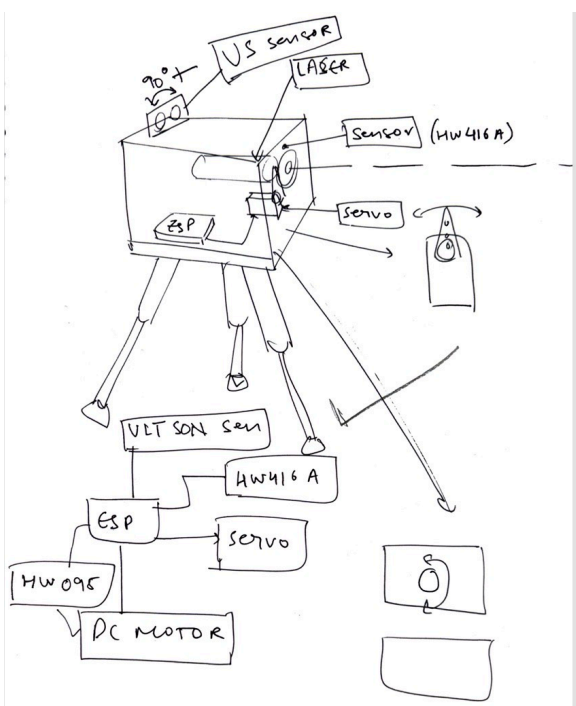
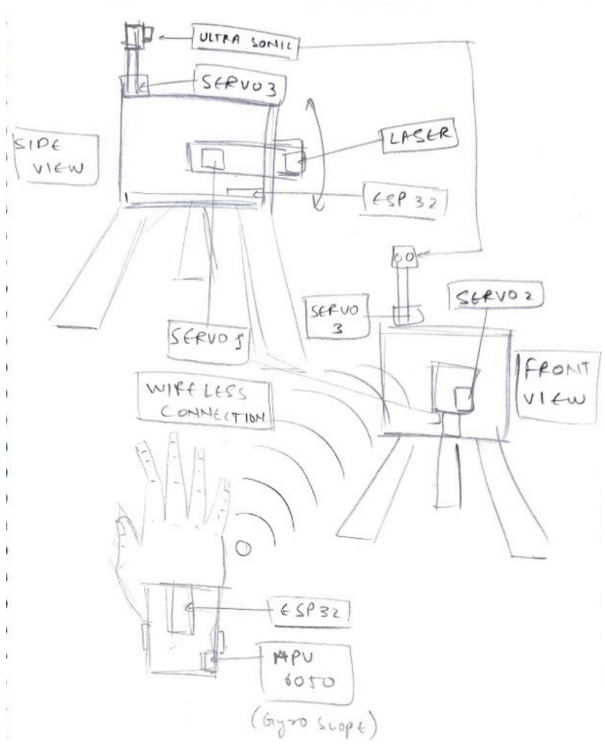
ODT Summative 2

Idea Brief :

GyroSight is a gesture-controlled, wireless turret system that explores how natural hand movements can be translated into precise mechanical action. Instead of traditional controls like buttons or joysticks, the system allows a user to aim a turret simply by tilting their hand creating a more intuitive and embodied interaction between human and machine.

The system consists of a handheld controller and a turret unit. The controller uses a motion sensor (MPU6050) to detect hand orientation and sends this data wirelessly via an ESP32 to the turret. The turret responds in real time using two servo motors for horizontal and vertical movement while a third servo rotates an ultrasonic sensor to continuously scan the surroundings. This enables the system to not only follow user input but also actively sense its environment.

When an object is detected within a certain range, a laser module activates to provide immediate visual feedback, simulating a targeting or alert mechanism. GyroSight brings together gesture-based control, environmental sensing and responsive feedback to create a cohesive interactive system highlighting how intuitive, body-driven interfaces can redefine the way users interact with physical devices.

Basic Idea Sketch :